MOHAMMAD HOSSEINI

Software Engineer | Backend Developer

Tehran, Iran | 09115219213 | mimhe1381@gmail.com
github.com/1mimhe | linkedin.com/in/mohammad-hosseini-17b953367

Technical Skills

Languages & Web: JavaScript, TypeScript, Node.js, HTML5, CSS3, EJS, Handlebars

Frameworks & APIs: NestJS, Express.js, GraphQL (Basics), RESTful APIs, Swagger

Databases & Data Layer: PostgreSQL, Redis, MongoDB, MySQL, Prisma, TypeORM, Sequelize, Mongoose

Architecture & Messaging: Event-Driven Architecture, Modular Monolith, Microservices (Basics), Message Brokers (RabbitMQ, Redis Streams), System Design

Real-Time Systems: WebSocket (Socket.IO)

DevOps & Tooling: Docker, Git, GitHub Actions, Amazon S3, Linux / Bash

Testing & Quality: Unit Testing (Jest, Vitest), Integration Testing (Supertest), API Testing

Professional Experience

Backend Developer — *Dec 2025 – Apr 2026*
Droplinked (droplinked.com) — *Abu Dhabi, UAE | Remote*

- Migrated legacy services to a modular NestJS architecture, refactoring core business logic for long-term maintainability.
- Designed and structured relational database schemas with indexing and data integrity constraints.
- Developed and deployed new product features end-to-end and resolved critical production bugs to improve system stability.

Node.js Intern — *Jun 2024 – Sep 2024*
Robintel (robintel.ir) — *Isfahan, Iran | On-Site*

- Built backend services with Express.js, implementing routing, middleware pipelines, and RESTful API endpoints following team conventions.
- Collaborated with the product team to design application flows and database schemas for new feature development.
- Practiced Git-based workflows including feature branching, pull requests, and code reviews in a team environment.

Projects

SyncBoard – Real-Time Collaborative Workspace — github.com/1mimhe/sync-board
TypeScript, NestJS, PostgreSQL, Prisma, Redis, RabbitMQ, Socket.IO, Yjs, Docker, Jest, Nginx

- Designed and built the backend of a Trello/Notion-style workspace (Kanban boards, cards, collaborative documents) as a NestJS modular monolith with decoupled domain modules communicating via domain events, ready for microservice extraction.
- Implemented real-time document editing with Yjs (CRDT) over Socket.IO with debounced PostgreSQL persistence and snapshot versioning; scaled WebSockets horizontally with the Redis adapter and built Redis-based presence tracking.
- Optimized PostgreSQL: monthly range-partitioned audit log, partial/composite indexes, and LexoRank ordering for single-row card reordering.
- Built async processing on RabbitMQ with dead letter exchanges, exponential-backoff retries, and Redis-backed idempotent consumers.
- Implemented RS256 JWT auth with rotating refresh tokens and reuse detection, Redis token blacklist, Google OAuth, multi-tenant RBAC, Redis rate limiting and idempotency keys, and presigned S3 uploads.
- Wrote 1,200+ unit tests (100% coverage), containerized the stack (Nginx, PgBouncer, read replicas), and built a React client to demo the backend.

NexCart – AI-Powered E-Commerce (In Progress)

- Learning and planning a microservices e-commerce backend around new concepts: Saga-based distributed transactions, Transactional Outbox with CDC and Kafka event streaming, gRPC contracts, OIDC authentication and API gateway, hybrid vector search with LLM function calling, and observability with contract and load testing.

HamBaar – Package Delivery Platform github.com/1mimhe/hambaar-backend
TypeScript, NestJS, PostgreSQL, Prisma, Redis, AWS S3, Docker, GitHub Actions, Jest, Turf.js

- Led product definition and backend design of a platform matching package senders with transporters, covering the full flow: package and trip creation, matching, delivery requests, tracking, and wallet with escrow and commission settlement.
- Built the matching engine using geospatial route-corridor analysis (Turf.js, Neshan Maps API) with capacity checks and ranking, plus a configurable dynamic pricing engine (distance tiers, weight, special handling, route deviation).
- Implemented multi-stage SMS OTP authentication with scoped JWTs (temporary, progress, access), RBAC, Redis sessions, OTP rate limiting, and presigned S3 uploads with size limits.
- Set up unit, integration (real PostgreSQL and Redis), and e2e tests (600+ unit tests, ~92% coverage) and a CI/CD pipeline deploying Docker images automatically.

MenuChi – No-Code Menu Builder github.com/1mimhe/menuchi-backend
TypeScript, Express.js, Prisma, PostgreSQL, Redis, AWS S3, TSOA, Docker, Vitest

- Built the backend of a no-code restaurant menu platform: a reusable item backlog and day-based menu publishing across a multi-tenant hierarchy (restaurant, branch, menu), plus customer ordering with status tracking.
- Implemented owner auth (session + JWT with scope-based permissions) and customer email-OTP login with rate limiting.
- Integrated external image-processing and OTP services via Redis Streams, and presigned S3 uploads for menu images.

Bookstore API github.com/1mimhe/bookstore-nest
TypeScript, NestJS, TypeORM, MySQL, Redis, Docker, Jest

- Designed a normalized MySQL schema modeling titles, physical editions, authors, translators, publishers, tags, characters, collections, orders, and more, with TypeORM migrations and seeders.
- Built the e-commerce flow: carts, percentage/fixed discount codes, and orders with price snapshots and atomic conditional stock updates to prevent overselling under concurrent requests.
- Implemented session-based auth (Redis) with role guards (customer, staff, publisher, admin), plus non-blocking Redis view tracking and trending content.
- Built reviews with replies and emoji reactions, bookmarks, blogs, and support tickets, with event-driven staff audit logging, health probes, and Swagger docs.

Education

Bachelor of Science in Computer Engineering *2021 – 2025*
University of Isfahan

Additional

Languages: English (Professional Working Proficiency) | Persian (Native)

Soft Skills: Team Collaboration, Ownership & Responsibility, Fast Learner, Adaptability